

Descriptive Statistics Project

MTH-155-42: Statistical Reasoning

Name: Cheyenne [REDACTED]

Date: 11/29/2025

Instructor: Professor Samuel Chukwuemeka

Project: Descriptive statistics of the average number of cardiac arrests encountered by EMS in the City of Austin, Texas per fiscal quarter from Oct. 13, 2014 to Mar. 19, 2019.

Dataset: Number of cardiac arrests encountered by EMS in the City of Austin, TX from 2014 Fiscal Quarter 1 to 2019 Fiscal Quarter 2.

Source of dataset: Data.gov. URL: <https://catalog.data.gov/dataset/ems-quarterly-cardiac-arrest-measures>

Instructions:

- (1.) Open link to webpage
- (2.) Scroll down to “Downloads & Resources”
- (3.) Download “Comma Separated Values File”
- (4.) Open download
- (5.) It should be viewable in Excel format

Below is an image of the Excel file and the data included in the dataset. The area lined in red is the dataset I will be analyzing. This data represents the count of cardiac arrests encountered by EMS in the City of Austin, TX, per fiscal quarter, from Oct. 13, 2014 to Mar. 19, 2019.

	A	B	C	D	E	F	G	H	I	J	K
1	Fiscal Qua	Fiscal Qua	Fiscal Qua	Fiscal Qua	Count â€“	Count â€“	Percent â€“	Target - Percent	Cardiac Arrests with Pulse at ED		
2	201401	2014-Q1	Oct-13	Dec-13	150	42	28	30			
3	201402	2014-Q2	Jan-14	Mar-14	159	52	32.7	30			
4	201403	2014-Q3	Apr-14	Jun-14	117	47	40.17	30			
5	201404	2014-Q4	Jul-14	Sep-14	116	30	25.86	30			
6	201501	2015-Q1	Oct-14	Dec-14	138	41	29.71	30			
7	201502	2015-Q2	Jan-15	Mar-15	135	41	30.37	30			
8	201503	2015-Q3	Apr-15	Jun-15	155	52	33.55	30			
9	201504	2015-Q4	Jul-15	Sep-15	128	33	25.78	30			
10	201601	2016-Q1	Oct-15	Dec-15	141	37	26.24	30			
11	201602	2016-Q2	Jan-16	Mar-16	138	41	29.71	30			
12	201603	2016-Q3	Apr-16	Jun-16	145	55	37.93	30			
13	201604	2016-Q4	Jul-16	Sep-16	158	52	32.91	30			
14	201701	2017-Q1	Oct-16	Dec-16	155	42	27.1	30			
15	201702	2017-Q2	Jan-17	Mar-17	159	46	28.93	30			
16	201703	2017-Q3	Apr-17	Jun-17	146	38	26.03	30			
17	201704	2017-Q4	Jul-17	Sep-17	145	38	26.21	30			
18	201801	2018-Q1	Oct-17	Dec-17	143	46	32.17	30			
19	201802	2018-Q2	Jan-18	Mar-18	155	35	22.58	30			
20	201803	2018-Q3	Apr-18	Jun-18	114	28	24.56	30			
21	201804	2018-Q4	Jul-18	Sep-18	144	43	29.86	30			
22	201901	2019-Q1	Oct-18	Dec-18	143	39	27.27	30			
23	201902	2019-Q2	Jan-19	Mar-19	96	30	31.25	30			

Values:

150, 159, 117, 116, 138, 135, 155, 128, 141, 138, 145, 158, 155, 159, 146, 145, 143, 155, 114, 144, 143, and 96

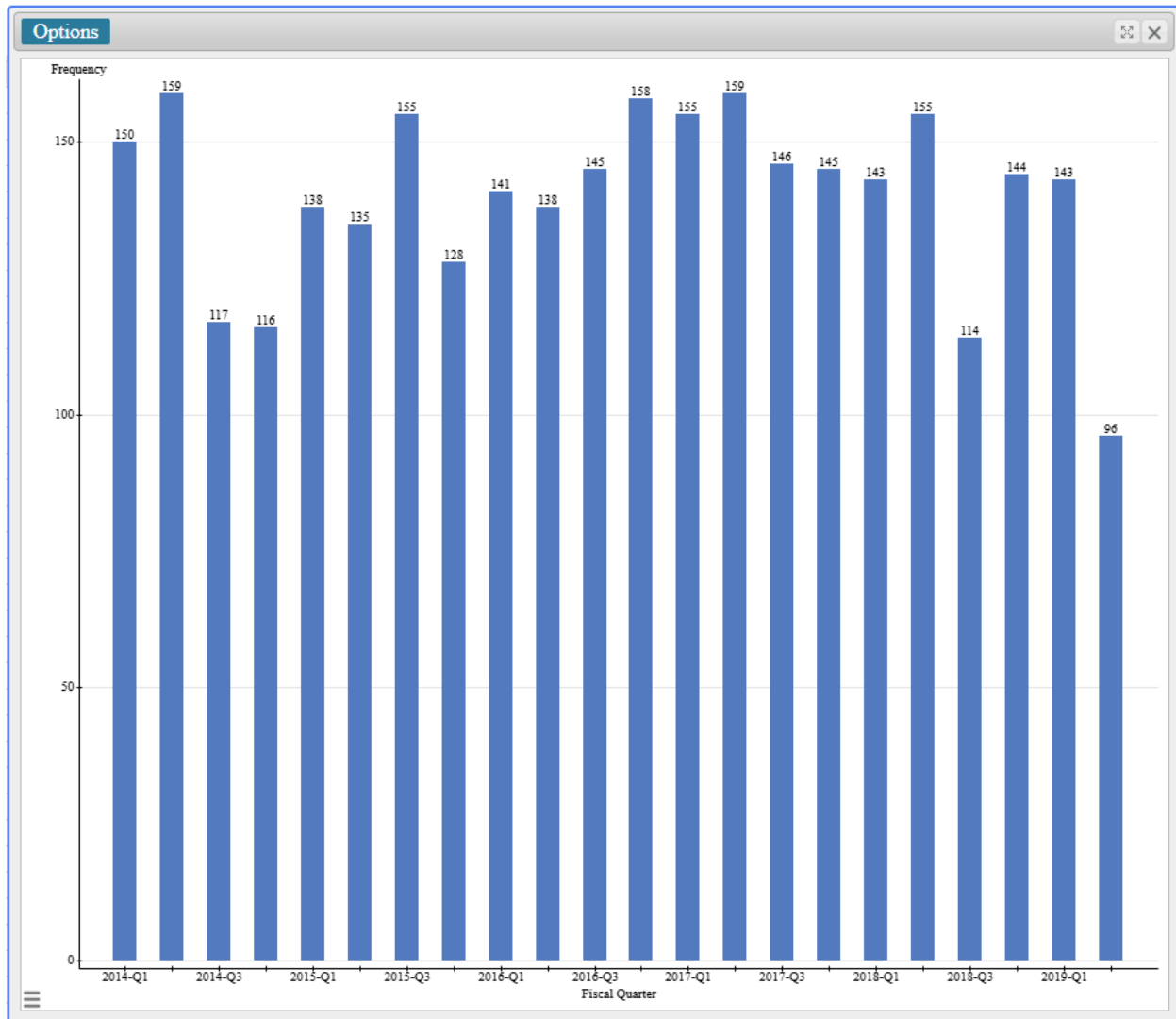
Objectives:

In this project, I will:

- (1.) Present the data in a bar chart.
- (2.) Analyze and describe the data.
- (3.) Calculate and interpret the measures of center within the context of the dataset.
- (4.) Calculate and interpret the measures of spread within the context of the dataset.
- (5.) Calculate and interpret the measures of position within the context of the dataset.

Technology Used:

I will use the TI-84 Plus calculator to verify my results as well as StatCrunch. I have also used StatCrunch to organize the data and create the bar chart.

Presentation:

This bar chart was generated using StatCrunch software and shows the number of cardiac arrests encountered by EMS in the City of Austin from Oct. 13, 2014 to Mar. 19, 2019. Values represent the number of these cardiac arrests that occurred each fiscal quarter. The quarter with the smallest count of cardiac arrests, 96

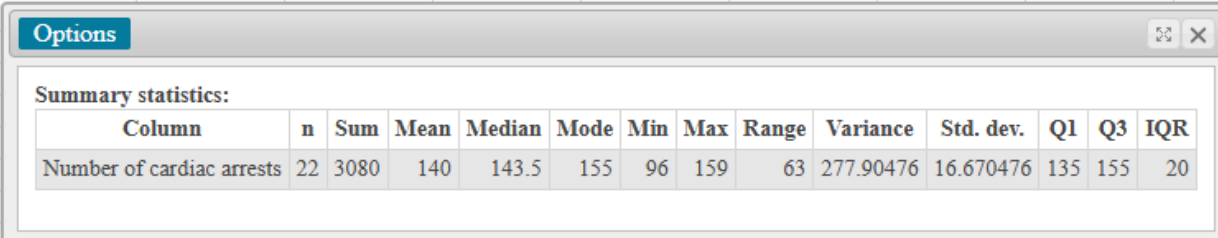
cardiac arrests, occurred in Fiscal Quarter 2 of 2019 from Jan. 19 – Mar. 19. Two quarters had the largest count of cardiac arrests, 159 cardiac arrests, which occurred in Fiscal Quarter 2 of 2014 from Jan. 14 – Mar. 14 and Fiscal Quarter 2 of 2017 from Jan. 17 – Mar. 17.

Analysis of the Data:

Analysis will include measures of center (mean, median, mode, and midrange)

Analysis will include measures of spread (range, variance, standard deviation, and interquartile range)

Analysis will include measures of position (five number summary).



The screenshot shows a software window titled "Options" with a close button. Inside, there is a section labeled "Summary statistics:" followed by a table of statistical data.

Column	n	Sum	Mean	Median	Mode	Min	Max	Range	Variance	Std. dev.	Q1	Q3	IQR
Number of cardiac arrests	22	3080	140	143.5	155	96	159	63	277.90476	16.670476	135	155	20

Let x represent the values for the cardiac arrests encountered by EMS in the City of Austin, Texas from Oct. 13, 2014 to Mar. 19, 2019.

Measures of Center:

Mean

$$\text{Mean} = \bar{x} = \frac{\sum x}{n}$$

$$\begin{aligned}\sum x &= (150 + 159 + 117 + 116 + 138 + 135 + 155 + 128 \\ &\quad + 141 + 138 + 145 + 158 + 155 + 159 + 146 \\ &\quad + 145 + 143 + 155 + 114 + 144 + 143 + 96) \\ &= 3080\end{aligned}$$

$$n = 22$$

$$\text{Mean}, \bar{x} = \frac{\sum x}{n} = 3080/22 = 140 \text{ cardiac arrests.}$$

The mean of the average cardiac arrests per fiscal quarter encountered by EMS in Austin, TX between Oct. 13, 2014 to Mar. 19, 2019 is 140 cardiac arrests.

Median

To calculate the median of the data, I will sort the data values in ascending order:

96, 114, 116, 117, 128, 135, 138, 138, 141, 143, 143, 144, 145, 145, 146, 150, 155, 155,

155, 158, 159, 159

$$\text{Median} = Q_2 = \frac{143 + 144}{2} = \frac{287}{2} = 143.5$$

When the number of cardiac arrests per fiscal quarter encountered by EMS in Austin, TX between Oct. 13, 2014 to Mar. 19, 2019 are arranged in ascending order, the median when rounded to a whole number is 144 cardiac arrests. The median is rounded because the data represents a discrete variable.

Mode

Three fiscal quarters in Austin, TX in the period from Oct. 13, 2014 to Mar. 19, 2019 had 155, making the mode of the dataset 155 cardiac arrests.

Midrange

$$\text{Midrange} = \frac{\text{minimum} + \text{maximum}}{2} = \frac{96 + 159}{2} = \frac{255}{2} = 127.5 \approx 128$$

The midrange of the number of cardiac arrests per fiscal quarter encountered by EMS in Austin, TX between Oct. 13, 2014 to Mar. 19, 2019 when

rounded to the nearest whole number is 128. The value is rounded to a whole number because the data are discrete variables.

Measures of Spread

Range

$$\text{Range} = \text{maximum} - \text{minimum}$$

$$159 - 96 = 63$$

The difference between the maximum and minimum number of cardiac arrests per fiscal quarter encountered by EMS in Austin, TX between Oct. 13, 2014 to Mar. 19, 2019 is 63.

Variance

$$\text{Variance} = s^2 = \frac{\sum(x - \bar{x})^2}{n - 1}$$

To calculate the variance, I will use a table.

Number of Cardiac arrests, x	$x - \bar{x}$	$(x - \bar{x})^2$
150	10	100
159	19	361
117	-23	529
116	-24	576
138	-2	4
135	-5	25
155	15	225
128	-12	144
141	1	1
138	-2	4
145	5	25
158	18	324
155	15	225
159	19	361
146	6	36
145	5	25
143	3	9

155	15	225
114	-26	676
144	4	16
143	3	9
96	-44	1936
$\Sigma(x - \bar{x})^2 = 5836$		

$$\text{Variance} = s^2 = \frac{\Sigma(x - \bar{x})^2}{n - 1} = \frac{5836}{22 - 1} = 277.9047619$$

The variance for the dataset is 277.9047619. This shows that there is a lot of spread in the number of cardiac arrests in Austin, TX for the dataset's time period.

Standard Deviation

The standard deviation is equal to the square root of the variance.

$$\begin{aligned}\text{Standard deviation, } s &= \sqrt{\text{Variance}} = \sqrt{277.9047619} \\ &= 16.67047576\end{aligned}$$

The standard deviation for the dataset is 16.67047576 cardiac arrests.

This is 16.67048 when rounded to 5 decimal places.

The standard deviation shows there is a considerable amount of spread amongst the number of cardiac arrests recorded per quarter in Austin, TX.

Interquartile Range

$$\text{Interquartile Range, } IQR = Q_3 - Q_1$$

$$Q_3 = \text{Upper Quartile}$$

$$Q_1 = \text{Lower Quartile}$$

To find the IQR, we must sort the data in ascending order:

96, 114, 116, 117, 128, 135, 138, 138, 141, 143, 143, 144, 145, 145, 146, 150, 155,
155, 155, 158, 159, 159

$$n = 22$$

$$Q_1 = \frac{1}{4} * n$$

$$\frac{1}{4} * 22 = \frac{11}{2} = 5.5 \approx 6$$

5.5 rounded up is 6. This value is rounded up because this is a discrete variable.

The Lower Quartile is the value in the 6th position, which is 135.

$$Q_3 = \frac{3}{4} * n$$

$$\frac{3}{4} * 22 = \frac{33}{2} = 16.5 \approx 17$$

16.5 rounded up is 17. This value is rounded up because this is a discrete variable.

The Upper Quartile is the value in the 17th position, which is 155.

$$Q_1 = 135$$

$$Q_3 = 155$$

$$IQR = Q_3 - Q_1 = 155 - 135 = 20$$

The interquartile range for this dataset is 20. The variation in 50% of the data on the number of cardiac arrests per fiscal quarter encountered by EMS in Austin, TX from Oct. 13 2014 to Mar. 19 2019 is 20 arrests.

Measures of Position

Five Number Summary

The Five Number Summary includes the minimum value, Q_1 , the median (Q_2), Q_3 , and the maximum value.

Minimum value for the dataset = 96

$$Q_1 = 135$$

$$\text{Median} = 143.5 \cong 144$$

$$Q_3 = 155$$

Maximum value for the dataset = 159

The Five Number Summary for the number of cardiac arrests per fiscal quarter encountered by EMS in Austin, TX from Oct. 13 2014 to Mar. 19 2019 shows that:

- (1.) 96 cardiac arrests is the minimum number of cardiac arrests per fiscal quarter encountered by EMS that occurred in Austin, TX from Oct. 13, 2014 to Mar. 19, 2019.
- (2.) 135 is the 25th percentile of cardiac arrests per fiscal quarter encountered by EMS in Austin, TX from Oct. 13, 2014 to Mar. 19, 2019. This means that 25% of the dataset is below 135 cardiac arrests.
- (3.) 143.5 is the median or 50th percentile of cardiac arrests per fiscal quarter encountered by EMS in Austin, TX from Oct. 13, 2014 to Mar. 19, 2019. This means that 50% of the dataset are below nearly 144 cardiac arrests.
- (4.) 155 is the 75th percentile of cardiac arrests per fiscal quarter encountered by EMS in Austin, TX from Oct. 13, 2014 to Mar. 19, 2019. This means that 75% of the dataset are below 155 cardiac arrests.
- (5.) 159 is the maximum number of cardiac arrests per fiscal quarter in Austin, TX from Oct. 13, 2014 to Mar. 19, 2019.

References (APA 7th Edition)

Chukwuemeka, S. (2019) *Descriptive Statistics*. Descriptive Statistics.

<https://statistical-science.appspot.com/DescriptiveStatistics/descriptive-statistics.html#>

City of Austin, Texas. (2025, April 25). *EMS- Quarterly Cardiac Arrest Measures*.

Data.gov. <https://catalog.data.gov/dataset/ems-quarterly-cardiac-arrest-measures>